

Railway track fastener defect detection based on image processing and deep learning techniques: A comparative study[☆]

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ABSTRACT

The railway track fasteners play a critical role in fixing the track on the ballast bed. Achieving full automation of the fastener defect detection is significant in terms of ensuring track safety, and reducing maintains cost. In this paper, innovative and intelligent methods using image processing technologies and deep learning networks are proposed. In the first part, the traditional fastener positioning method based on image processing is reconsidered. In addition, a novel fastener defect detection and identification method using Dense-SIFT features is proposed which can achieve a better performance than the methods available in the literature. In the second part, VGG16 is trained for fastener defect detection and recognition. The result demonstrates that it is possible to carry out the defect detection of fasteners with CNN. Finally, Faster R-CNN is used for fastener defect detection to advance detection rate and efficiency. The fastener positioning and recognition can be carried out simultaneously. The time for the defect detection and classification is only one-tenth of the other methods mentioned above.

1. Introduction

In the last two decades, as one of the most important modern modes of transportation, rail transit has been greatly developed all over the world, especially in China. The rapid development of rail transportation puts forward more strict requirements on transportation safety. The health state of rails and fasteners is critical to ensure the safe and stable operation of rail transit. Owing to the effect of the contact friction and vibration impact between train wheels and track, coupled with the influence of the natural environment, defects such as broken or missing fasteners are likely to occur on track lines. The track fasteners are applied to fix the track on the ballast bed. While some missing fasteners or partly broken fasteners occur in the track line, part of the track cannot be fixed tightly on the ballast bed. The track will often be deformed, displaced gradually and even collapsed due to the strong impact force while trains run across the track day and night. Therefore, it is critically important to detect the missing and broken, or partly worn fasteners in time. Nowadays, the inspection task for these track components is carried out by trained engineers who walk along the track line and check the health state of the fasteners periodically. Manual inspection is a tough work that is time consuming, inefficient and costly. It is urgent to develop track fastener detection system for improving the safety of the railway transit, inspecting the track automatically, shortening

the inspection time, reducing the maintenance cost and alleviating the labour intensity of the workers by using the well developed technologies such as image processing, computer vision, machine learning, and fast speed and high resolution cameras (Zuwen, 2007).

In the last decade, many researchers and institutions have devoted their efforts to the development of automatic inspection methods and systems for railway. The automatic detection of the railway tunnels by a robot not only improves the detection efficiency, but also reduces the risk of detecting workers (Montero et al., 2015). The defect management systems based on image-matching and augmented reality (AR) enables workers and managers to detect dimension errors and omissions automatically on the job site (Kwon et al., 2014). ENSCO's RIS (Railway imaging systems) has advanced high-resolution image acquisition systems and image processing algorithms, which can continuously detect fasteners during the day and night (Berry et al., 2008). The bvSys company's RailCheck system can detect the loss of rail-based components such as fasteners at a speed of 200 km/h (BvSys, 2018). However, the system is only used by local companies. The details of technologies are not reported. Besides the existing prototype system equipment, some researchers have also studied railway track fastener detection techniques based on image processing in the laboratory environment. In Marino et al. (2007) and De Ruvo et al. (2009), multilayered

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perception neural classifier is used to detect missing fasteners, and the algorithm is implemented for online fastener detection with the help of GPU. In Yang et al. (2011), LDA (Blei et al., 2003) is used to produce weight coefficient matrix based on the direction field of fasteners and then the matching distance is calculated by the fastener image and the coefficient matrix for missing fastener detection. However, the method does not consider the detection of partially broken fasteners. In Xia et al. (2010), template matching method is used to detect the fastener regions. Then the Haar-like feature that extracted from fastener regions is used as the input of the AdaBoost algorithm. However, the normal fastener template cannot match the broken or missing fastener. Furthermore, the Haar-like feature has the disadvantage of weak feature representation ability, even though its advantage is the simple calculation. In Jiajia et al. (2016), the fastener location is detected by empirical value, in which the track position is almost fixed. But the algorithm lacks robustness to the track position variation. In Li et al. (2014) and Li et al. (2011), the geometric features of the rail components are used to position fasteners, where the visible horizontal lines in rail track are detected by Hough transform and Sobel edge detection. In Resendiz et al. (2013), 2-D images are transformed into 1-D signals by Gabor filter, and then the multiple signal classification (MUSIC) algorithm is used to detect the 1-D signals, which can classify the signals produced by different track component. But the fasteners that mentioned in these three papers are very different from the fasteners considered in this paper which are widely used in Metro track lines. In Feng et al. (2014), the position of the rail and sleeper is fixed by LDA algorithm. The fastener position is located by using the positional relationship among fasteners, sleepers and track. The Haar-like feature extracted from single fastener image is used to compute the likelihood probabilities for defects classification. However, the average classification precision of missing fastener in the paper needs to be improved. In Resendiz et al. (2013), the HOG features of fastener image are applied to SVM (Support Vector Machine) for classification. But the recall performance of the method for missing fastener and broken fastener classification is not good enough for practical application yet. In addition, feature extraction methods used in these papers such as Gabor (Wang et al., 2010; Gibert et al., 2015), Edge detection (Yitzhaky and Peli, 2003), and Hough transform (Bowen et al., 2015) ignore the extraction of local features in the images. The fasteners are composed of different components. It is quite important to extract the local features for defect identification. With the development of deep learning theory, the research of DCNN (Deep Convolutional Neural Networks) has been achieved significant results, particularly in the area of image classification (Krizhevsky et al., 2012; Szegedy et al., 2015) and object detection (Girshick et al., 2014). In Makantasis et al. (2015), CNN is used for tunnel inspection. In comparison with the other methods such as SVM, LDS, and etc., CNN has great advantages in detection accuracy and detection rate. In Gibert et al. (2017), MTL (Multitask Learning) is used for railway tie and fastener detection. Eight categories of material are classified first. Based on the material classification result, three kinds of the fastener are classified. However, the paper does not discuss the detection rate of the Multitask detector. In Loupos et al. (2018), Faster R-CNN is used for crack detection. However the detection method are susceptible to noise. There is no other reports on the deep learning method for track fastener identification yet.

The techniques of fastener defect detection have achieved significant progress. However, there are still some problems needed to be solved. The positioning accuracy and robustness are the first two problems need to be considered. Improved fastener defect feature extraction needs to be developed so that better classification accuracy can be achieved. Finally, the time consuming for the whole procedure and each single image should be shortened so that the improved method can be applied to the online real-time practical application.

Aiming at these three main problems, in this paper, new methods are proposed to determine the boundaries of fasteners with the help of variance projection and wavelet transform. The positioning accuracy

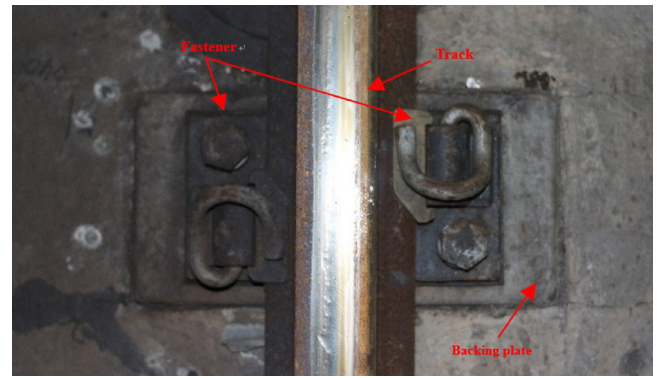


Fig. 1. The fastener used for Beijing Metro Line 6.

is improved since there is no extra content in the fastener area being captured. In the fastener classification part, Dense-SIFT is used to extracting local features of fasteners, and the accuracy of defective fastener detection is improved by using Bag-of-Visual-Word and spatial pyramid decomposition techniques. Besides, DCNN (e.g., VGG16 Simonyan and Zisserman, 2014) is proposed to classify the different statues of fastener defects. Objects in the image are segmented by locating their regions and determining their sizes in the image. After that, they are recognized by the CNN. In our case, there are two different objects in our images, the track and two fasteners. We localize the fastener and cut them out from the images and then feed them into the CNN. To improve the detection speed for real-time defect detection, we propose to use the object detection network, Faster R-CNN (Ren et al., 2015), for fastener detection. RCNN(region-CNN) can carry out positioning and classification together in one network. It consists of a regional proposal sub-network and a CNN classifier. R-CNN is a classical network in the field of object detection. Fast-RCNN and faster-RCNN are two new improved versions of R-CNN. The method of deep learning is more robust than the traditional image recognition algorithm. To the best of our knowledge, this is the first research that introduces Faster R-CNN to the fastener detection and classification.

This paper is organized as follows. In Section 2, the problems solved in this paper are stated. In the third section, the track fastener positioning issue is investigated, and several methods are proposed. In Section 4, the fastener identification based on Dense-SIFT and SVM is provided. A CNN, more specifically, VGG16 based fastener classification is presented in Section 5. After that, a fast objection detection method based on Faster R-CNN is investigated in Section 6. Finally, some conclusions are given in Section 7.

2. Track fastener and problem statement

The track fastener is the component used to connect the track and sleeper on rail transit lines, which ensures that the rail and sleeper are relatively fixed. Moreover, it can reduce the deformation of the track due to its elasticity. Nowadays, most of the metro lines use concrete rail bed instead of ballast or sleepers. In this paper, a type of track fastener in Beijing Metro Line 6 is used as an example for the fastener defect detection and classification issue. One image containing a track, two fasteners, and the backing plate is shown in Fig. 1.

According to the on-site investigation of the Beijing Metro Line 6 and the information obtained from maintenance engineers. The defects of the track fastener mainly consist of two categories, broken fastener and missing fastener, as shown in Fig. 2(a) and Fig. 2(b), respectively. The broken fastener is defined as the elastic bar fracture or partly worn fastener. The missing fastener is defined as the main component loss or the fastener lost totally.

There are two critical steps for fastener defect detection. The first step is to position the location of the fastener in the images precisely

which are photographed from the railway track field. The second step is to isolate the three different fastener defects (complete, broken and missing) in the light of the defect features.

Fastener positioning: The positioning algorithm should be useful for different detective fasteners. As it is shown in Fig. 1, the main components in the image consist of two fasteners, track and rail bed. Fasteners are installed on both sides of the track. Fastener positioning is achieved by searching the boundaries of the track and the concrete rail bed. In this paper, innovative methods will be proposed for the positioning issue so that the performance of classification accuracy and robustness can be met.

Fastener classification based on Dense-SIFT: The defects of the fasteners are resulted from the elastic bar fracture of the fastener or missing of the main fastener components. In consideration of the spatial relationship of fastener parts, the features extract method used in this paper is Dense-SIFT based on spatial pyramid decomposition. The features extracted from fastener images are fed into SVM for classification.

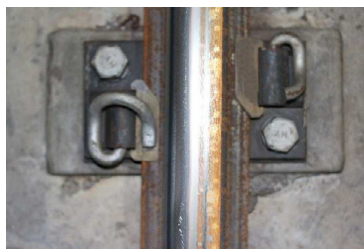
Fastener classification based on DCNN: To simplify the fastener detection and classification procedure, the deep learning theory, more specifically DCNN, is applied to the fastener detection and classification issue. The network used in this part is VGG16 which include 13 convolution layers and 3 fully connected layers. The proposed DCNN-based method uses only one deep neural network for the considered issue. The parameters need to be tuned are much fewer than those of the approach based on image feature extraction.

Fast fastener classification based on Faster R-CNN To accelerate the calculation speed and reduce the time consuming on each image, in this paper, the recently developed object detection algorithm, Faster R-CNN (Ren et al., 2015), is introduced to achieve the real-time detection. This algorithm can carry out both positioning and classification simultaneously.

The dataset used in this paper is made up of images that are taken from Beijing Metro Line 6. Fig. 3 shows the tunnel where the images are collected. The images are captured by handheld DSLR camera. The camera angle is perpendicular to the fastener, and the distance between the camera and fastener is roughly the same. These images are RGB images with a resolution of 3468×5472 pixels. The collected images are characterized by the track in the middle of the image, and each image contains at least two fasteners. We have collected 322 available images from the Nanluoguxiang Station to the Dongsi Station of Beijing Metro Line 6.

3. Fastener positioning based on image processing

The fastener, backing plate and track are the three main parts of the metro line. There is a fixed spatial relationships among these three parts. It is difficult to positioning fasteners directly. The commonly used method is to segment the fastener from the other parts by taking advantage of the spatial relation of the track, the fastener and the backing plate. Based on the image of railway containing fasteners collected at Beijing Metro Line 6, it is observed that these fastener images have three obvious characteristics:



(a) Broken fastener



(b) Missing fastener

Fig. 2. Defects of fasteners.



Fig. 3. Data collection tunnel at Beijing Metro Line 6.

1. The texture feature of fastener is different from the other parts.
2. The width and height of the fastener do not exceed the width and height of the backing plate, respectively.
3. The brightness of the fastener is lower than that of the track and the backing plate.

Fig. 4 shows that the grey value of the track surface area is almost invariant. The grey value of other areas has clear change, especially at the edges. In the light of these characteristics, the area of the track and backing plate can be split firstly by using image processing techniques. The fastener boundaries that need to be located in this paper are marked with red lines as shown in Fig. 4.

3.1. Image preprocessing

Owing to the influence of interferences such as the dusty environment and camera shake, it is easy to generate noised images and lower the quality image. This will affect the image positioning, identification and the accuracy of detection results. The noise needs to be filtered in advance. In this paper, median filter (Hodgson et al., 1985; Weiss, 2006) is applied to eliminate noise while protecting the defect edge information.

$$f(x, y) = \text{median}_{(s,t) \in S_{xy}} \{g(s, t)\} \quad (1)$$

where $f(x, y)$ is pixel value of the processed image. x, y stand the coordinate of each pixel. $g(s, t)$ is the template of median filter, the size is 3×3 .

3.2. Track and backing plate positioning

Image binarization can reduce the impact of the other interferences on the image quality and also reduce the amount of calculation for further processing. In this paper, grey scale image is transformed into

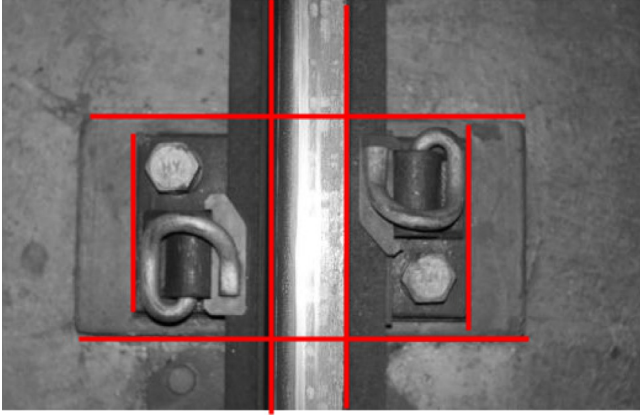


Fig. 4. Grey image of fastener.

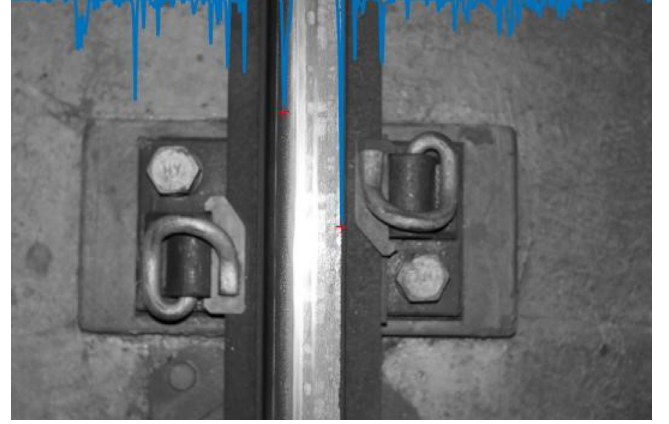


Fig. 6. The result of vertical projection.

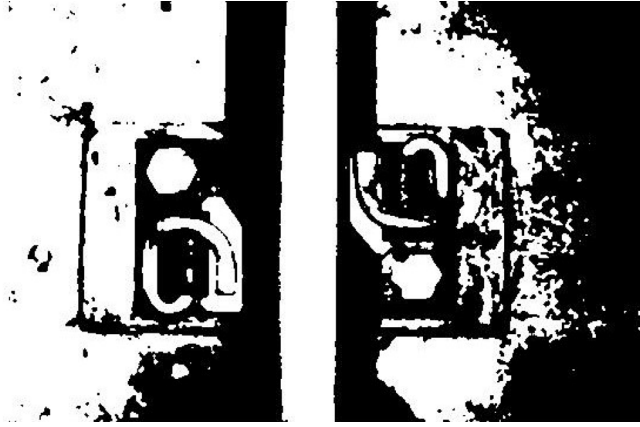


Fig. 5. The binary image of fastener.

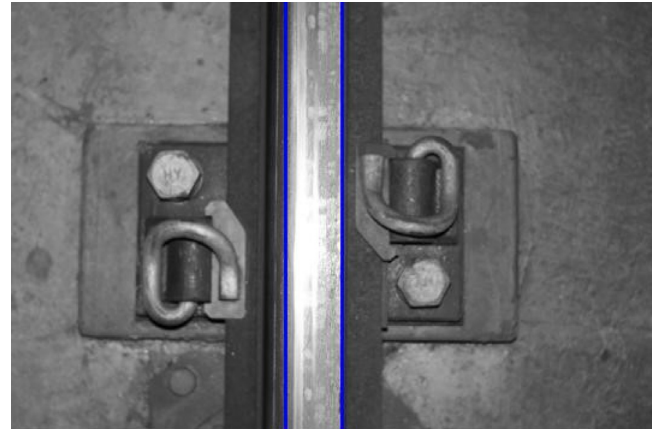


Fig. 7. The track edge position result.

a binary image by adopting a method based on maximum between-cluster variance (Otsu, 1979). This method can meet the demand of the automatic threshold segmentation of fastener images. The binary image obtained with this method is shown in Fig. 5.

After that, the binary image is processed by a vertical projection as follows

$$VP(j) = \sum_{i=1}^h \text{Binary}f(i, j), i = 1, 2, \dots, w, j = 1, 2, \dots, h \quad (2)$$

where $VP(i, j)$ is the result of the vertical projection, $\text{Binary}f(i, j)$ is the pixel value of the binary image, which means $\text{Binary}f(i, j)$ equals 0 or 1. The difference between the two adjacent VP is further calculated

$$DVP(j) = VP(i+1) - VP(i), i = 1, 2, \dots, w \quad (3)$$

and the obtained DVP is shown in Fig. 6 (the blue line in the image). In the light of the three image characteristics mentioned before, the DVP values of the two edges of the track should be the maximal points in the DVP (the two maximal points are marked by red '+'). After the two maximal points are determined, the two track edges can be fixed by two vertical lines. The track edge positioning result is shown in Fig. 7.

After positioning the track edge, we need to position the backing plate edge. The idea used for backing plate positioning and track positioning is the same. The only difference is the direction of the projection. The vertical projection is used in track positioning and, instead, the horizontal projection is used for backing plate positioning. However, the horizontal boundary of the backing plate is not as clear as the track boundary. To validate the effectiveness of the proposed algorithm, 13 images are used to determine the backing plate boundary

and also calculate their widths by using the projection algorithm. The determined widths of the backing plate are shown in Fig. 8.

It can be seen that there are several backing plate widths do not match the actual widths (e.g. frame 2, frame 4, frame 5, frame 11, frame 12) which is caused by inaccurate positioning as one example shown in Fig. 9(a). This indicates that the projection method is not effective and its robustness is not good enough. To overcome this problem and enhance the robustness of the algorithm, the projection algorithm is improved by adding the cache of edge array (Babenko, 2009). The backing plate width of each frame of the image is stored in the cache. If the backing plate width detected in this frame does not match predefined range, this detected result is ignored. The algorithm will search for a maximum value within 10 pixels around near the position that obtained from the previous frame detection result. The principle of the improved algorithm is as follows

$$\text{edge} = \text{argmax}_i DVP(i), C_{pre} - ed \leq i \leq C_{pre} + ed \quad (4)$$

where edge is the position of backing plate edge. ed is the searching range, which is obtained by multiple experiments and set to 10 in this paper.

The improved positioning results for the backing plate are shown in Fig. 9(b). Compared it with Fig. 9(a), the improved algorithm accurately positions the horizontal boundary of the backing plate. The same 13 images as mentioned before are used to verify the accuracy of the improved algorithm, and the result is shown in Fig. 10. It can be seen that the widths of the backing plates are estimated with much higher accuracy in comparison with the projection algorithm.

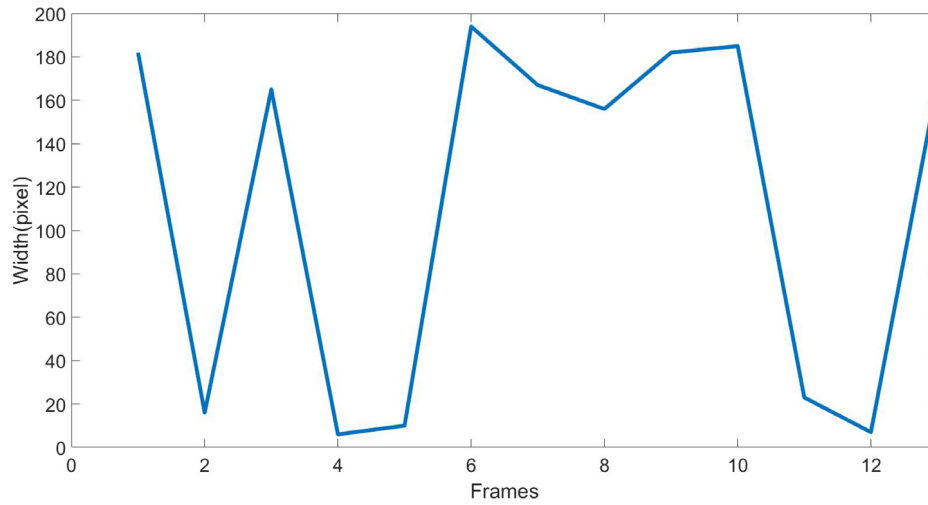
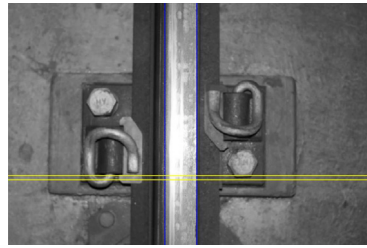
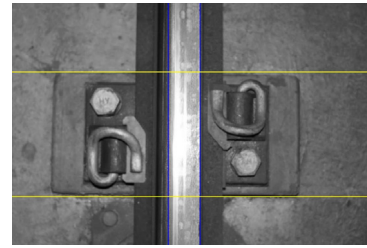


Fig. 8. The estimated backing plate widths by using horizontal projection difference method.



(a) The estimated boundaries by using the projection algorithm



(b) The estimated boundaries by using the improved projection algorithm

Fig. 9. The estimated backing plate boundaries.

3.3. Fastener positioning

In the last two subsections, the track edges and the edges of the backing plate are determined by the projection algorithm. To detect the fastener defects, it is necessary to further determine the target boundary of the fastener as shown in Fig. 11. In this paper, this edge is positioned by using db4 wavelet transform.

Wavelet transform uses wavelets to represent a signal or a function. The discrete wavelet is defined as follows

$$\Psi_{j,k} = |a_0|^{-\frac{j}{2}} \Psi(a_0^{-j}t - kb_0), j, k \in \mathbb{Z} \quad (5)$$

$$DW_{j,k} = \int_R f(t) \Psi_{j,k}^*(t) dt \quad (6)$$

where $\Psi_{j,k}(t)$ stands for the mother wavelet based on the parameters a and b . $f(t)$ is the function of $L^2(R)$. In general, $a_0 = 2$ and $b_0 = 1$. Generally, an image is processed by 2-D discrete wavelet transform, which means the image is performed by discrete 1-D wavelet transform from the horizontal direction and the vertical direction. Mallat algorithm in 2-dimension is used in image decomposition and reconstruction (Niya and Aghagolzadeh, 2004). In a certain scale, let the low frequency component be a_i , high frequency components be d_i , the low-pass filter be h , the high-pass filter be g . Then the 2-D decomposition as follows.

$$a_{j+1}(n_1, n_2) = \sum_{k_1 \in \mathbb{Z}} \sum_{k_2 \in \mathbb{Z}} h_{m_1-2n_1} h_{m_2-2n_2} a_j(m_1, m_2) \quad (7)$$

$$d_{j+1}^1(n_1, n_2) = \sum_{k_1 \in \mathbb{Z}} \sum_{k_2 \in \mathbb{Z}} h_{m_1-2n_1} g_{m_2-2n_2} a_j(m_1, m_2) \quad (8)$$

$$d_{j+1}^2(n_1, n_2) = \sum_{k_1 \in \mathbb{Z}} \sum_{k_2 \in \mathbb{Z}} g_{m_1-2n_1} h_{m_2-2n_2} a_j(m_1, m_2) \quad (9)$$

$$d_{j+1}^3(n_1, n_2) = \sum_{k_1 \in \mathbb{Z}} \sum_{k_2 \in \mathbb{Z}} g_{m_1-2n_1} g_{m_2-2n_2} a_j(m_1, m_2) \quad (10)$$

The synthesis filter is h and g , then the reconstruction as follows.

$$\begin{aligned} a_j(n_1, n_2) = & \sum_{m_1 \in \mathbb{Z}} \sum_{m_2 \in \mathbb{Z}} h_{m_1-2n_1} h_{m_2-2n_2} a_{j+1}(n_1, n_2) \\ & + \sum_{m_1 \in \mathbb{Z}} \sum_{m_2 \in \mathbb{Z}} h_{m_1-2n_1} g_{m_2-2n_2} d_{j+1}^1(n_1, n_2) \\ & + \sum_{m_1 \in \mathbb{Z}} \sum_{m_2 \in \mathbb{Z}} g_{m_1-2n_1} h_{m_2-2n_2} d_{j+1}^2(n_1, n_2) \\ & + \sum_{k_1 \in \mathbb{Z}} \sum_{k_2 \in \mathbb{Z}} g_{m_1-2n_1} g_{m_2-2n_2} d_{j+1}^3(n_1, n_2) \end{aligned}$$

$j = 0, 1, 2, \dots$

Since the target boundary is vertical, the positioning method is based on a vertical component of the wavelet transform. The vertical wavelet transform result is shown in Fig. 12(a). It can be seen that the image is divided into three main parts, track and the two fasteners on the left and right side of the track, respectively. There are still some interferences and noise in the processed image. A unit of line structure is designed for morphological opening operation to filter the noise and interferences. The angle of the unit and the target boundary is the same. The image filtered by the morphological opening operation (Gonzalez et al., 2009) is shown in Fig. 12(b).

After reducing the noise in the image, an operator template is designed for positioning the target boundary. The statistical distribution of template matching is as follows

$$D(i, j) = \sum_{h=1}^H \sum_{w=1}^W [S_{ij}(h, w) \cap f_{ij}(h, w)] \quad (11)$$

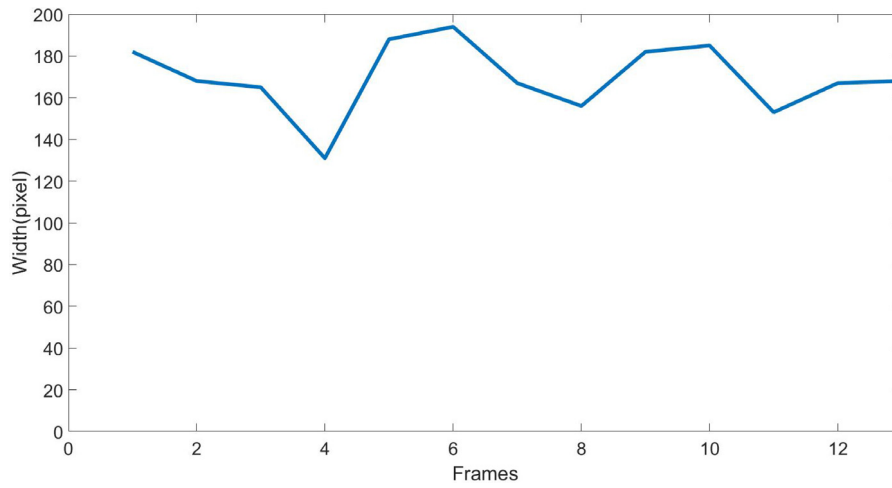


Fig. 10. The estimated backing plate widths by using the improved algorithm.

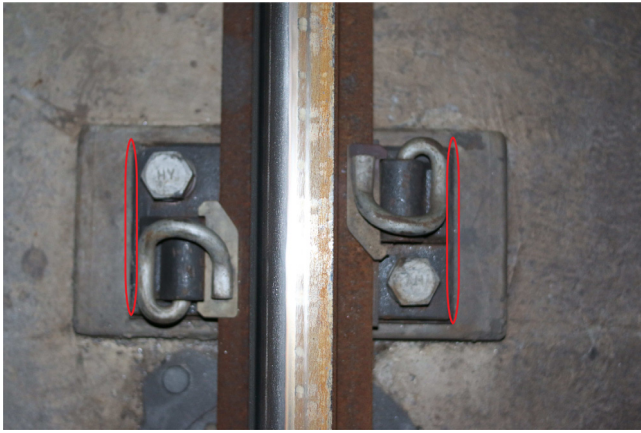


Fig. 11. The target boundary need to detection.

where S is the template, f is the image of vertical component, $\cap$ stands for *and* operation. h and w stand for the coordinates of each pixel. $D(i, j)$ stands for the matching degree.

Based on the track boundary positioning results, the pixel value outside the width of the half track boundary is set to 0. The reason is two fold. On the one hand, it can partly eliminate the interference of the vertical edge of the track for matching. On the other hand, it can reduce the amount of the template matching calculation. After that, a template matching operator is carried out. The template size is 91×21 , which is obtained from multiple experiments. The result of template matching is shown in Fig. 13(a). The higher the matching degree, the whiter it is

in the image. Finally, the target boundaries are determined as shown in Fig. 13(b).

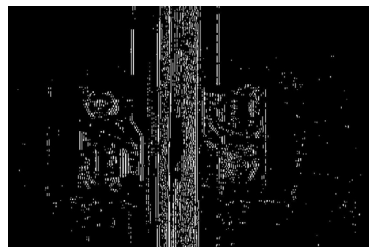
By taking all the positioning means for the track, the backing plate and also the fastener boundary together, the final fastener positioning result is shown in Fig. 14. It can be seen that the two fastener boundaries are determined precisely. The fastener positioning method proposed in this paper is robust and does not require the fastener position to be fixed in the image. This method achieves better positioning results than those methods reported in the literature.

4. Fastener classification based on image feature extraction

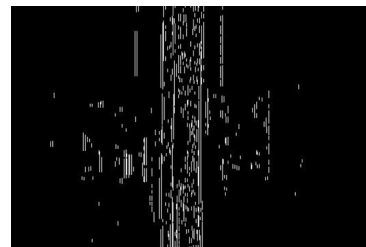
In the preceding section, the problem of fastener positioning is solved, and the resulted image data set only contains the fastener without the track and the backing plate. In the following, the fastener detection algorithm based on the local feature extraction and classifier will be presented first. The popular algorithm for local features extraction is Dense Scale Invariant Feature Transform (Dense-SIFT) (Fei-Fei and Perona, 2005). A new method is proposed in this paper that constructs Bag-of-visual-word model (Kato and Harada, 2014) of the fastener feature based on Dense-SIFT first, and then the features are fed to SVM for fastener classification.

4.1. Dense SIFT feature (Fei-Fei and Perona, 2005)

The procedure for extracting local features based on Dense-SIFT is shown in Fig. 15. The main function of the Dense-SIFT algorithm is to extract the image feature that is not sensitive to the variation of illumination, scale, rotation, and etc., which is robust to the geometric transformation of the image, too. The disadvantage of the SIFT algorithm lacks some global features which probably lead to misclassification



(a) Inverse transformation of vertical detail



(b) Result after noise removing

Fig. 12. The Inverse transformation of vertical component of wavelet transform.

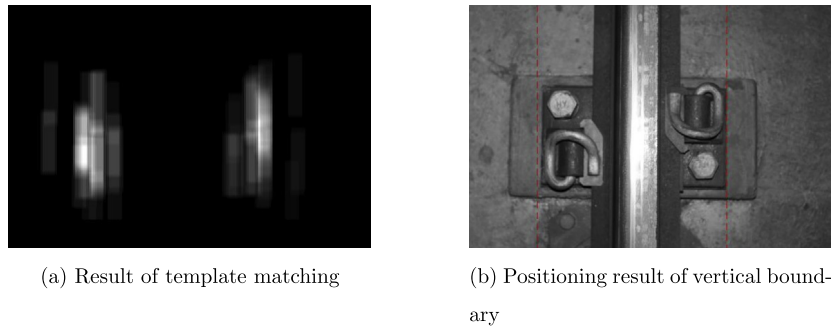


Fig. 13. The positioning result of vertical boundary.

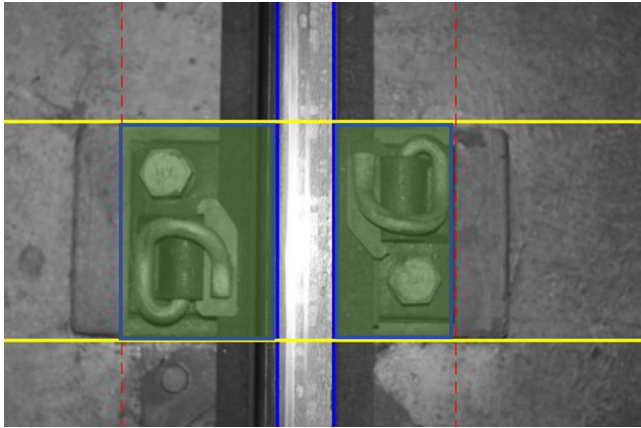


Fig. 14. The positioning result of fastener.

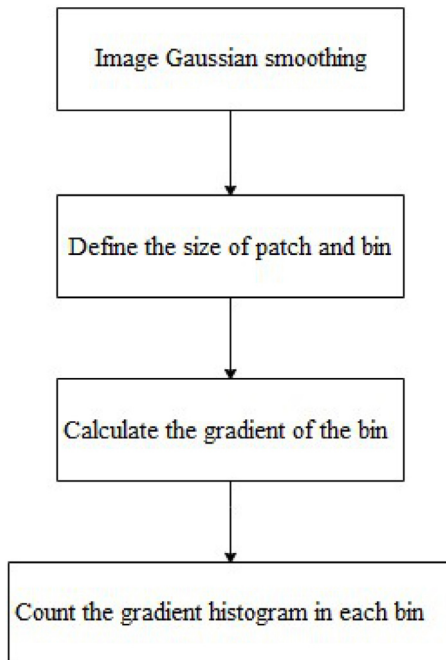


Fig. 15. The calculating process of Dense SIFT feature.

in different scenes. In comparison with SIFT algorithm (Lowe, 1999), global features are supplemented by Dense-SIFT algorithm (Fei-Fei and Perona, 2005), which can improve the object classification accuracy in different scenarios.

The key of Dense-SIFT algorithm is to determine the descriptors of each bin. In this paper, the method of determining the descriptors is to calculate the gradient magnitude and the direction of each bin, which are introduced in Lowe (2004). They are reviewed here briefly.

$$m(x, y) = \sqrt{(L(x+1, y) - L(x-1, y))^2 + (L(x, y+1) - L(x, y-1))^2} \quad (12)$$

$$\theta(x, y) = \arctan\left(\frac{L(x, y+1) - L(x, y-1)}{L(x+1, y) - L(x-1, y)}\right) \quad (13)$$

where $L(x, y)$ stands the input image. $m(x, y)$ is the gradient magnitude of each bin. $\theta(x, y)$ is the gradient direction of each bin. In this paper, the patch size is 4 bins $\times$ 4 bins and the bin size is 4 pixels $\times$ 4 pixels. The sampling step is set as 8 pixels. the Dense-SIFT features of keypoints is shown in Fig. 16.

The yellow circle in Fig. 16 is the keypoint and the green boxes indicate a description of the Dense SIFT features extracted at this point. The more obvious edges presented in the region are, the larger the gradient modulus of the SIFT feature description are. On the contrary, the gradient modulus are small if the grey values in the region are homogeneous. More details about Dense-SIFT algorithm can be found in Fei-Fei and Perona (2005) and Lowe (2004).

4.2. Bag-of-visual-word model

Bag-of-Visual-Word (BOVW) (Kato and Harada, 2014) is developed from BOW (Bag-of-Word) (Salton et al., 1975). BOVW regards an image as a document and regards pixels or features in the image as words. In this paper, the model is applied to the fastener classification by constructing the bag of visual words of the fastener. The procedure for constructing a bag of visual word includes feature extraction and characteristic dictionary construction. Finally, an image feature description based on the statistical histogram is obtained. The procedure is shown in Fig. 17.

The critical step of constructing a bag of visual word is to map Dense-SIFT features extracted from each image to visual feature words, and then build these words into a visual characteristic dictionary. This process is achieved by K-means clustering. More details about BOVW can be found in Kato and Harada (2014).

4.3. Spatial pyramid decomposition

BOVW can extract the local feature of the image very well. However, it lacks the spatial location information of the image. To overcome this disadvantage, spatial pyramid decomposition (Lazebnik et al., 2006) of the image is introduced. The main idea of spatial pyramid decomposition is that the image is divided into several regions by a grid of N layers. In the N th layer of the grid, the x and y directions of the image are divided into 2^N equal-sized cells. Hence the number of sub-region is 4^N . For each sub-regions, the frequency that all visual words in the visual feature dictionary appearing in the region are counted. Therefore, each sub-region can generate a frequency statistical histogram based

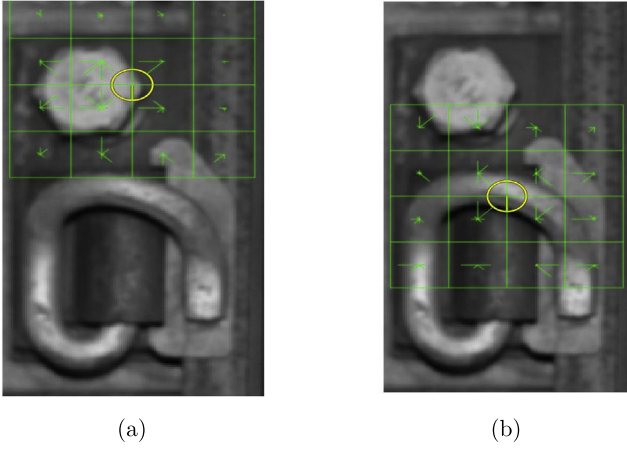


Fig. 16. Dense SIFT feature of the key points in a fastener image.

on visual feature words. For the different grid of layers, the frequency statistical histogram of all the sub-regions is arranged in order, which constitutes the feature vector of the layer based on the visual word dictionary.

$$H_x^N = (H_x^{N,1}, H_x^{N,2}, \dots, H_x^{N,k}, \dots, H_x^{N,4^N}) \quad (14)$$

where H_x^N stands for the feature vector based on visual feature words of image x in the N th layer. $H_x^{N,k}$ stands for the frequency statistical histogram of the k th sub-region. Fig. 18 is the diagram of visual characteristic word bag model based on spatial pyramid decomposition. The image to be analysed is decomposed into 1, 2 and 3 layers. Then a frequency statistical histogram of each sub-region by counting the frequency of each visual feature word is obtained. The icons of different shapes represent each visual word in the visual feature dictionary.

4.4. Training process

The three types of fasteners used in the training experiment are shown in Fig. 19. We ended up with 574 complete fasteners and 20 defective fasteners, including 13 broken fasteners and 7 missing fasteners. Due to the small number of defective fasteners, data augmentation methods are used to extend the defective samples. The data augmentation methods used in this paper include mirror, flip, rotation and adding noise. A sample of data augmentation is shown in Fig. 20, in which (a) is the original image, (b) is the flipped image, (c) is the

Table 1

The number of different fasteners in the dataset.

Types	Complete	Broken	Missing
Training	105	105	105
Testing	45	45	45

Table 2

Pyramid decomposition accuracy comparison.

	Spatial pyramid decomposition	Without Spatial pyramid decomposition
Accuracy	99.26%	96.30%

mirror image, (d) is an image with 5% salt noise, (e) is an image added Gaussian noise with a mean of 0.1 and a variance of 0.01.

After data augmentation, the data set of fastener contains 574 complete fasteners, 201 broken fasteners and 150 missing fasteners. Since unbalanced samples can affect the classification results, sub-sampling is used for complete fasteners and broken fasteners samples. The data used for experiment contains 150 samples for each type. 70% of the images are used for training, and the rest of 30% of the images are used for testing. The data set is shown in Table 1 and Fig. 19.

In this paper, the fastener images used are resized to 128 pixels $\times$ 256 pixels and smoothed with a Gaussian operator with variance of 0.5. The patch size is 4 bins $\times$ 4 bins, and the bin size is 4 pixels $\times$ 4 pixels, the step of the sampling area is 8 pixels. After Dense-SIFT feature extraction, we construct the bag of visual word with a dictionary size of 10 by K-means clustering. Each image is decomposed by spatial pyramid decomposition with 3 layers. Too few layers will result in the loss of features in each cell, and too many layers can cause the feature dimension to be too high. Finally, the extracted visual feature is fed to SVM with a kernel of RBF.

A simulation experiment is carried out to verify whether pyramid decomposition has an impact on fastener classification or not, and the result is shown in Table 2. It turns out that pyramid decomposition improves the classification accuracy by 3%.

To demonstrate the effect of dictionary sizes to the classification accuracy, some experiments under different dictionary sizes are carried out, and the results are shown in Table 3. It can be seen from Table 3, the detection accuracy does not change significantly when the dictionary sizes larger than 10. On the contrary, when the dictionary size is larger than 20, the detection accuracy decreased. Therefore, the optimal dictionary size of BOVW is 10 for this case.

Table 4 is the confusion matrix of classification results under the different states of the fasteners. The confusion matrix shows detailed classification results for each type of fastener.

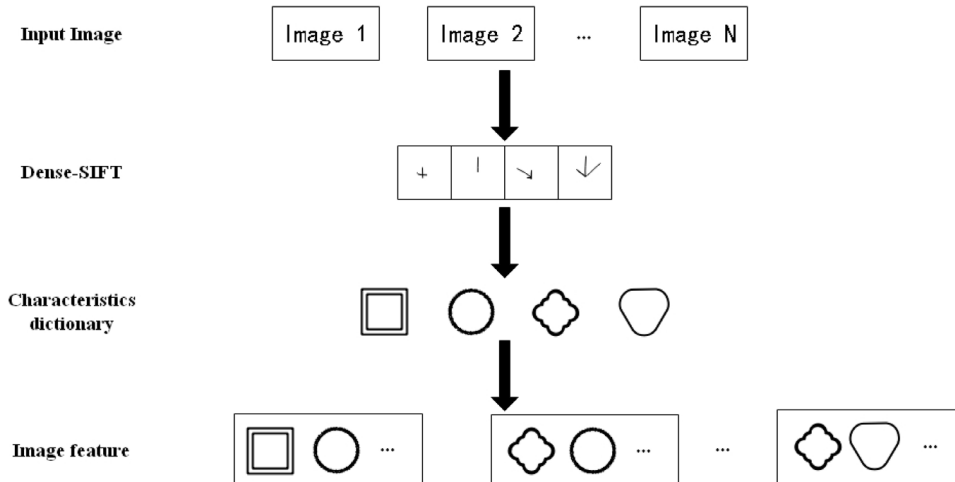


Fig. 17. The procedure of BOVW construction.

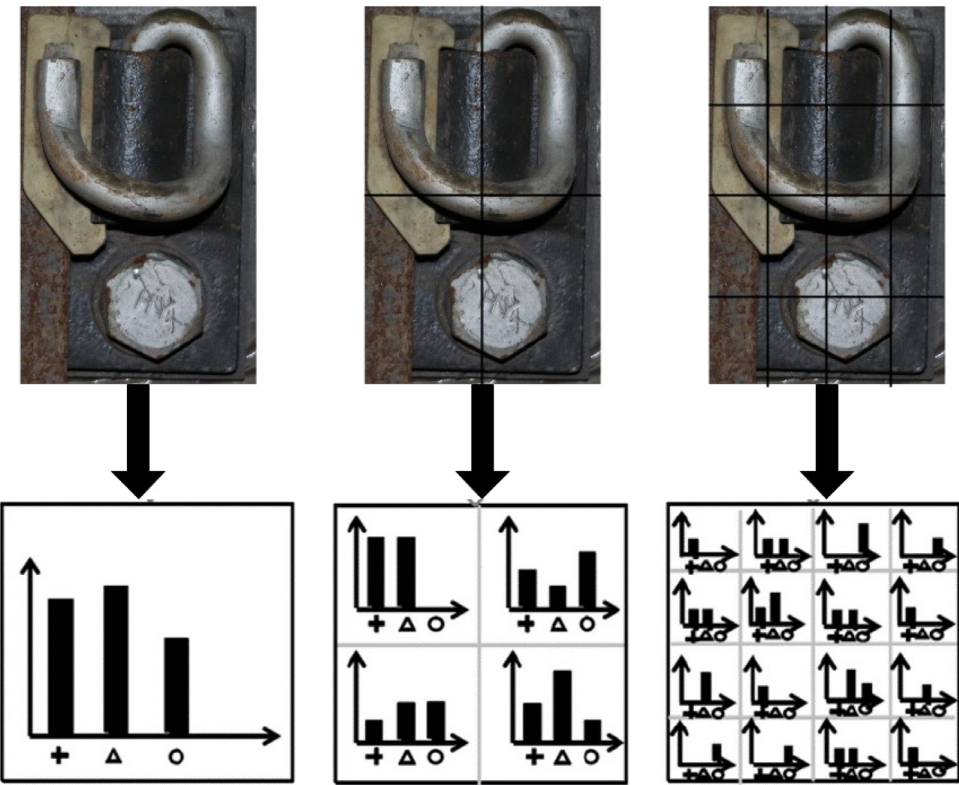


Fig. 18. The diagram of visual characteristic word bag model based on Spatial Pyramid decomposition.



Fig. 19. Some fastener dataset used in this paper.

Table 3
The comparison results under different dictionary size with SVM.

Dictionary size	Accuracy
5	96.29%
10	99.26%
15	99.26%
20	98.51%

The precision, recall and F1 score are essential indicators for evaluating the quality of the classifier. The formulae for calculating these indicators are defined by formulae (15)–(17), respectively. A value close to 1 for these indicators indicates a better classification effect. Table 5 provides the classification result for these three indicators. It can be seen that the algorithm proposed in this paper achieves a sound performance. The precision and recall in the recognition complete fastener achieve

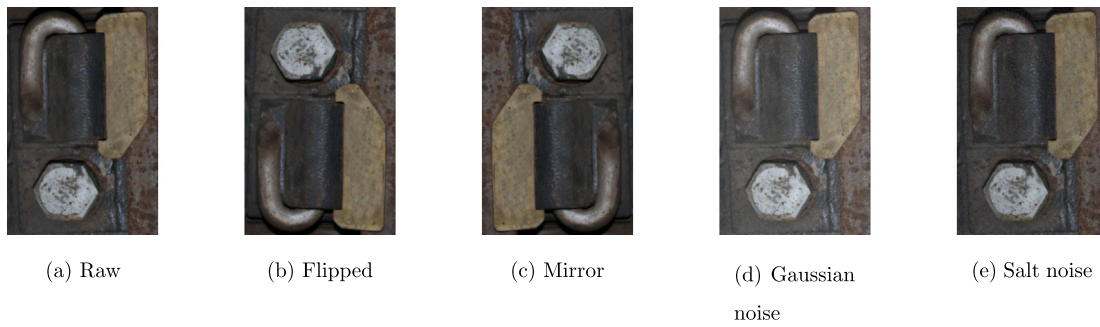


Fig. 20. The results of data augmentation.

Table 4

The confusion matrix of fastener detection based on image process.

Actual class	Predicted class		
	Complete	Broken	Missing
Complete	45	0	0
Broken	0	45	1
Missing	0	0	44

Table 5

Evaluation result for fastener state classification base on pyramid decomposition.

Type of fastener	Recall	Precision	F1
Complete	100%	100%	1
Broken	100%	97.67%	0.9882
Missing	97.87%	100%	0.9892

Table 6

Classification accuracy of different method.

Method of feature extraction	Accuracy
LBP	0.9407
HOG	0.9556
This paper	0.9926

100%.

$$P = \frac{TP}{TP + FP} \quad (15)$$

$$R = \frac{TP}{TP + FN} \quad (16)$$

$$F1 = \frac{2 \times P \times R}{P + R} \quad (17)$$

To demonstrate the effectiveness of the method proposed in this paper, a comparison is made for the other feature extractions methods including LBP feature and HOG feature. The result is shown in Table 6.

The method proposed in this paper achieves an accuracy of 99.26% which outperforms the other methods reported in the literature. The reason for this result is that the LBP feature and HOG feature ignore the spatial positional relationship of fastener features, which leads to the misclassification. In this paper, the spatial pyramid decomposition is used to extract the feature distribution in the image, which achieves a higher precision for the fastener defect detection.

5. Fastener defect detection and classification based on DCNN

In the last section, the fastener defect detection and classification method based on Dense-SIFT are presented. The final classification result achieves a very high precision. However, there are many steps and many techniques applied in the method, and there are also many parameters in the method need to be tuned for achieving a sound classification performance. Overall, it is a complicated method to classify fastener

images by using image feature extraction and classifier. It is expected that there is a method which has fewer steps and is easier to be used for practical application. In the last decades, the Deep Learning theory is developed very well. Some deep learning networks are designed for image recognition and classification, which are not sensitive to the scale, rotation, illumination and even robust to noises. In this section, inspired by deep learning theory, the well-developed DCNN (Deep Convolutional Neural Network which is a deep feed forward network that has been successfully used for image recognition (Krizhevsky et al., 2012)) is used for fastener classification. The deep learning network can fuse the low-level features of data, and complete the abstraction and extraction of advanced features of complex data. It is possible to build a deep network with the ability for feature extraction and image classification. Therefore, DCNN has some advantages over the image feature based method proposed in the last section. The DCNN structure used in this paper is VGG16 reported in Simonyan and Zisserman (2014).

5.1. Preprocessing of the fastener image

The image data set used for training VGG16 network is shown in Table 7. The image is an RGB colour image with three channels instead of the grey image with one channel. All the images are labelled with number 0,1,2, and then each label is encoded with one-hot encoding (Harris and Harris, 2010). One-hot encoding is a common method for categorical data. CNN cannot operate on label data directly. It requires all output variables to be transformed into numerical value. One-hot encoding can convert the integer value to orthogonal vectors as the label data.

In addition, normalization is used for all the images. It is useful for improving the classification accuracy and accelerating the convergence speed while training the network. The images are normalized by the following formula

$$x' = \frac{x - \text{mean}}{\text{stddev}} \quad (18)$$

where x is the pixel value of raw image on R, B, G channel. *Mean* is the average of all values in the raw image and *stddev* is the standard deviation of all values in the raw image.

5.2. The VGG16 architecture

The network architecture is shown in Table 8. In the table, *Conv* stands for the convolutional layer. *Max Pool* means max pooling layer. *Fc* stands for fully connected layer.

There are 14 total convolutional layers, and the kernel size is 3×3 in each convolutional layer. The number of convolution kernel with the network layer increases from shallow to deep gradually. Using multiple convolution kernels with small size instead of a single large convolution kernel can reduce the number of parameters and computation, and improve the nonlinearity of the network. The network uses rectified linear unit (ReLU) as the activation functions for all the layers. The method of pool layer is max pooling, and the unit size is 2×2 .

Table 7

Cross reference of image labels, defect class and dataset.

Image label	One-hot encoding	Defect category	Training set	Testing set	Validation set
0	[1,0,0]	Complete	105	45	35
1	[0,1,0]	Broken	105	45	35
2	[0,0,1]	Missing	105	45	35

Table 8

VGG16 network architecture.

Input(128 * 256 RGB image)			
Name	Kernel size	Stride	Trainable
conv1_1	3 × 3 × 64	1 × 1 × 1 × 1	FALSE
conv1_2	3 × 3 × 64	1 × 1 × 1 × 1	FALSE
max pool 1	1 × 2 × 2 × 1	1 × 2 × 2 × 1	
conv2_1	3 × 3 × 128	1 × 1 × 1 × 1	FALSE
conv2_2	3 × 3 × 128	1 × 1 × 1 × 1	FALSE
max pool 2	1 × 2 × 2 × 1	1 × 2 × 2 × 1	
conv3_1	3 × 3 × 256	1 × 1 × 1 × 1	FALSE
conv3_2	3 × 3 × 256	1 × 1 × 1 × 1	FALSE
conv3_3	3 × 3 × 256	1 × 1 × 1 × 1	FALSE
max pool 3	1 × 2 × 2 × 1	1 × 2 × 2 × 1	
conv4_1	3 × 3 × 512	1 × 1 × 1 × 1	FALSE
conv4_2	3 × 3 × 512	1 × 1 × 1 × 1	FALSE
conv4_3	3 × 3 × 512	1 × 1 × 1 × 1	FALSE
max pool 4	1 × 2 × 2 × 1	1 × 2 × 2 × 1	
conv5_1	3 × 3 × 512	1 × 1 × 1 × 1	TRUE
conv5_2	3 × 3 × 512	1 × 1 × 1 × 1	TRUE
conv5_3	3 × 3 × 512	1 × 1 × 1 × 1	TRUE
max pool 5	1 × 2 × 2 × 1	1 × 2 × 2 × 1	
Fc 6	4096		TRUE
Fc 7	4096		TRUE
Fc 8	3		TRUE

The fully connected network, at last, contains two hidden layers with 4096 neurons. The role of the fully connected layers is to receive the high-dimensional features extracted by the convolutional layer and classify them into three categories. We use dropout (Hinton et al., 2012) regularization (with a ratio of 0.5) and L2 regularization (with a ratio of 0.0005) on Fc 6 and Fc 7 to avoid over-fitting during the training procedure.

Since the number of samples available for training is not large, and if we train the weighting parameters of each layer from zero or some random numbers, the trained network cannot achieve a sound result. We use the idea of transfer learning (Geng et al., 2016) that the weights trained based on other data sets are used for the weight values initialization in this paper. The pre-trained weights used in this paper are trained by using ImageNet (Deng, 2009), which are loaded into our model and fine-tuned the model using our fastener dataset. The detail of fine-tuning is that the weights of *conv1* to *conv4* are fixed and trained without our data set. The weights of *conv5* and the fully connected layers are updated only during the training process.

5.3. Experimental results

The VGG16 network is implemented in GOOGLE Tensorflow framework. The GPU used for this experiment is NVIDIA 1070ti. A total of 450 fastener images are applied for this experiment. 315 images are used for training the model, and the rest 135 images are used for testing the accuracy and generalization ability of the trained model. The detail of the data set is shown in Table 1 and Fig. 19.

The raw images have large image size, such as 900 × 1800 pixels or larger. Since the raw images (after the fastener position) have different sizes. Feeding these images into the network directly result in inconsistent feature dimensions. They need to be resized to the same size. Furthermore, the images need to be re-scaled to an appropriate size

Table 9

Training time and performance for different image scales.

Scales	Training time (s)	Accuracy
32 × 64	1742	89.52%
64 × 128	1866	92.38%
128 × 256	3248	95.23%
256 × 512	4410	60.70%

Table 10

The classification results under different network hyperparameters.

	Learning rate	Weight decay	Accuracy (Training set)	Accuracy (Testing set)
1	0.001	None	100%	83.7%
2	0.01	0.0005	62.61%	49.62%
3	0.001	0.0005	99.42%	95.23%
4	0.0001	0.0005	75.67%	65.18%

for meeting the classification accuracy on the one hand. The training time is also in an acceptable range on the other hand. To explore the impact of image size on network training time and performance, the validation set are resized to four scales, 32×64, 64 × 128, 128×256 and 256 × 512, respectively. The number of training iterations is set to 10 000. The training time and accuracy in validation set is shown in Table 9. The larger image scales have also been tested in the experiment. However, the training cannot be carried out due to hardware limitations.

As the image size increases, more training time is required. The raw images have large image size, which needs CNN with too many inputs and a larger number of weights in the rest of layers in CNN, and the number of convolution operations increases during training. Low-resolution images contain less information, which results in too little effective information for the extracted feature maps and cannot achieve better classification results. On the contrary, the high-resolution images results in high dimensional features, which contains some invalid features. Besides, the sizes of the fastener images obtained by the method of fastener positioning are not entirely consistent. They need to be corrected so that they can be fed to the CNN with same dimensions. These reasons lead to inaccurate classification results. The image size of 128 × 256 is suitable for fastener defect detection.

For demonstrating the effect of the network parameters to the detection accuracy, some experiments are carried out under different network hyper parameters. The samples for the validation set and the test set are different samples. The results are shown in Table 10. It can be seen from this table, experiment 1 has high training accuracy, but the accuracy of testing is not as good as the training set since the network is over-fitted. To reduce over-fitting, the weight decay with a ratio of 0.0005 is added to the network. Besides, the network is trained with three different learning rates. Too large learning rates can make the network unable to converge, and a low learning rate will cause the network to converge on a non-optimal solution. Finally, the learning rate with a ratio of 0.001 and weight decay with a ratio of 0.0005 are used in VGG-16 training procedure, and these parameters are also used in Faster R-CNN which will be introduced in Section 6. In addition, the batch size is set to 16, and the number of iterations is set to 15 000.

The graph of the loss function during the training procedure is shown in Fig. 21. The red line is the loss function value of the training set, and the blue line is the loss change of the test set. In the training of the first 2000 iterations, the loss function converges rapidly. After that, the value of loss function begins to decline gradually. The trained model performs very well on both the training set and the test set. The last

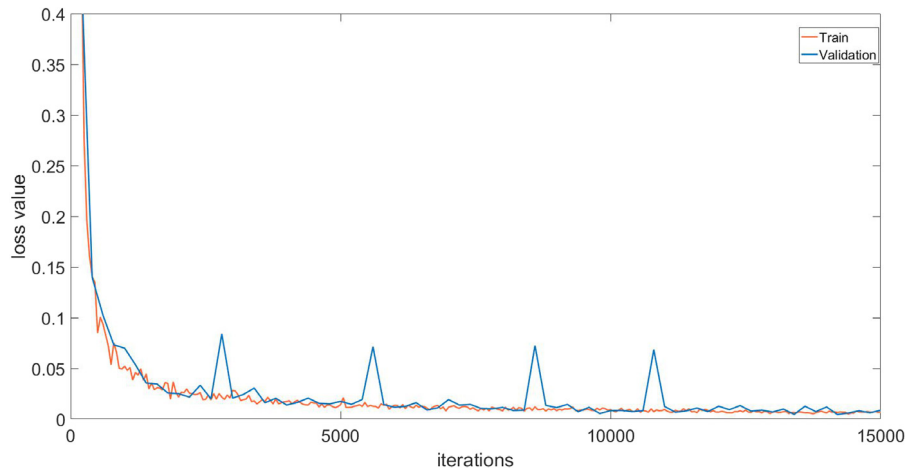


Fig. 21. The graph of the loss function during training based on VGG16. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 11

The confusion matrix of fastener detection based on VGG16.

Actual class	Predicted class			Recall
	Complete	Broken	Missing	
Complete	42	3	0	93.34%
Broken	0	45	0	100%
Missing	0	0	45	100%
Precision	100%	93.75%	100%	

trained Faster VGG16 network achieves an accuracy of 97.14% for the test set with 135 images.

The confusion matrix of classification result is shown in Table 11. It can be seen from the table that three complete fasteners are misclassified. In comparison with the classification results in Section 4, the accuracy on the test set dropped by 2% (99.26% to 97.14%). Since three complete fasteners are misclassified to broken fasteners, the precision of broken fastener is 93.75%. One misclassified fastener is selected and shown in Fig. 22. The reason for misclassification may be that the shadows block some of the fasteners, which leads the features extracted by this model is more similar to the feature of broken fasteners. The recall ratio reached 100% in the inspection of defective fasteners, which means all the defective fasteners are detected.

In summary, the accuracy of the fastener classification based on deep learning technology achieves 97.14%, which proves that it is very possible to carry out the defect detection of the fastener with deep learning theory. Due to the size of the data set, the advantages of the VGG16 network compared with traditional detection methods have not been performed and functioned fully. With the improvement of data acquisition equipment and computer hardware, more fastener defect image samples are obtained from the field, the DCNN network will be trained with enough images, and the classification accuracy will be improved significantly. This method can be applied to practical railway track fastener defect detection and classification tasks in the future.

6. Fastener defect detection and classification based on faster R-CNN

The DCNN based fastener detection and classification proposed in Section 5 combines the feature extraction and defect classification into one single network and achieves sound performance. However, the training procedure is slow, and this could be improved. With the development of deep learning, CNN can be used for region-based positioning and segmentation of objects. In the last few years, faster R-CNN (Ren et al., 2015) has been proposed, which has achieved better accuracy and faster processing rate. In this section, a brief introduction

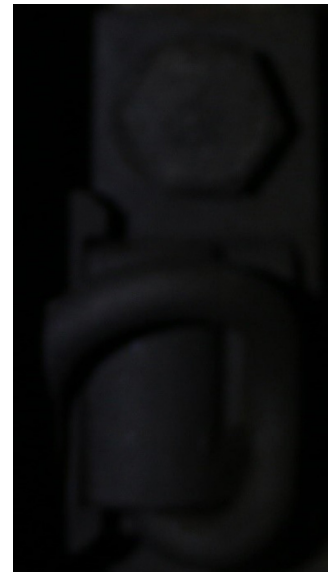


Fig. 22. The misclassified fastener.

about faster R-CNN will be made first, please refer to paper Ren et al. (2015) for more details. After that, it is applied to the fastener defect classification issue.

6.1. Faster R-CNN

Faster R-CNN is a deep learning network improved from Fast R-CNN (Girshick, 2015). And the main improvement is that RPN (Region Proposal Networks) is used for extracting the multiple proposal regions in the raw image. In the RPN, the feature extraction network (e.g. VGG16) is followed by a convolution layer in which the kernel size is 3×3 . This part is used for extracting features from the input image. A feature map containing the image information is obtained. Two 1×1 convolutional layers are applied for classification and bounding box regression (see Fig. 23).

For extracting the proposal region, anchors are introduced in RPN. First of all, the feature points are mapped back to the input image by the coordinate mapping. Each point of feature map corresponds to a field on the input image, which is called the receptive field. Then k anchors are set with different scale and aspect ratio. The default setting of k is 9. The three scales are 128^2 , 256^2 and 512^2 , respectively. The

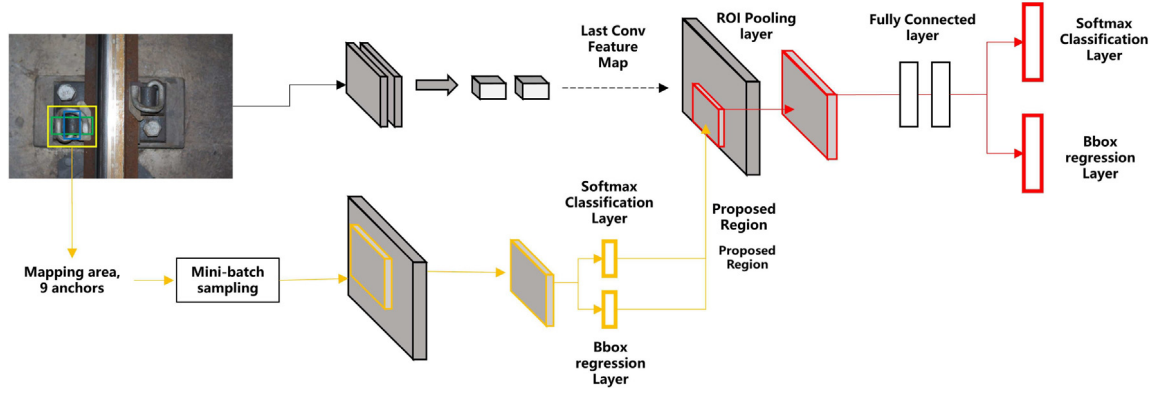


Fig. 23. Faster R-CNN architecture.

three aspect ratios are 1 : 1, 1 : 2 and 2 : 1, respectively. For the feature map size of $M \times N$, the number of proposal regions is $M \times N \times k$. Finally, after calculating the IOU (Intersection over Union) of proposal region and ground true box, the proposal region is labelled as positive samples or negative samples according to a set threshold. However, not all samples are involved in training. The number of negative samples is much larger than the number of positive samples. This will bias the trained network towards negative samples as they are dominated. To eliminate this impact, we randomly sample 256 anchors in each image and try to ensure that the number of positive and negative samples is equivalent. The RPN is trained end-to-end by stochastic gradient descent (SGD).

The training method of the network is 4-Step Alternating Training (Ren et al., 2015). In the first step, the RPN is initialized with a pre-trained model. In this paper, the weights used for initialization are the network parameters that obtained after training in Section 5. In the second step, the Fast R-CNN is trained by using the proposal regions generated by Step 1, which is also initialized by the weights that obtained after training in Section 5. These two networks do not share the convolutional layers. In the third step, the RPN is initialized by Fast R-CNN in Step 2, but the share convolutional layers are fixed and fine-tune the unique layer to RPN. Now the two networks share the convolutional layers. In the final step, the fully connected layers are fine-tuned with the convolutional layers fixed.

6.2. Image augmentation

Images used for object detection are large-scale images containing multiple object. In order to identify images containing multiple objects by Faster R-CNN, the single fastener image used in Sections 4 and 5 are not suitable for object detection experiment. The image data set used in this experiment is described in Section 2, which are RGB images with a resolution of 3468×5472 pixels.

The data augmentation method is used to increase the amount of data. The problem often encountered in anomaly detection is a small amount of abnormal data. The imbalance of the normal data and abnormal data in the dataset could bias the classification results of the trained network. The image augmentation is made including rotation, mirror and flipping. In addition, to overcome the problem of insufficient illumination of the image captured in the metro line, illumination augmentation is made based on Single Scale Retinex algorithm (Jobson et al., 1997) and the calculation formula is as follows

$$R_i(x, y) = \log S_i(x, y) - \log[F(x, y) * S_i(x, y)] \quad (19)$$

where the $R_i(x, y)$ is the retinex output. $S_i(x, y)$ represents a colour component image. $i \in R, G, B$. $F(x, y)$ is the surround function. $*$ stands the convolution operator.

The surround function is described as follows

$$F(x, y) = \lambda e^{-\frac{(x^2+y^2)}{c^2}} \quad (20)$$

Table 12

The detail of data set for experiment.

	Image	Complete	Broken	Missing
Training set	740	1159	236	68
Test set	318	507	95	32

Table 13

The confusion matrix of fastener detection based on faster R-CNN.

Actual class	Predicted class			Not detected	Recall
	Complete	Broken	Missing		
Complete	487	0	0	20	96.06%
Broken	2	93.00	0	0	98%
Missing	0	0	32	0	100%
Precision	100%	100%	100%		

Table 14

AP and mAP of fastener detection.

Class of fastener	AP	mAP
Complete fastener	0.948	
Broken fastener	0.989	0.979
Missing fastener	1	

Table 15

The computational cost of algorithms.

Algorithm	Time complexity	Space complexity
CNN(VGG16)	16.4×10^9	98.6M
Faster R-CNN	46.7×10^9	101.4M

where c is the scale of Gaussian surround. $\lambda = 1$ is selected so that

$$\iint F(x, y) dx dy = 1 \quad (21)$$

The result of illumination augmentation is shown in Fig. 24.

We end up with 1058 fastener images containing 1666 complete fasteners, 331 broken fasteners and 100 missing fasteners. 70% of the images are divided into training set and the rest 30% of the images as the test set. The data set for training and test is shown in Table 12.

6.3. Experimental results

The experimental environment is the same as the one introduced in Section 4. Batch size of proposal regions is set to 256. The learning rate is set to 0.001. The weight decay is set to 0.0005 and the number of iterations is set to 10 000. The threshold of IOU is selected as 0.5. The loss of changes in training procedure are shown in Fig. 25. In the figure, the loss class is the log loss of classification layer. The loss box is the loss value of Bbox regression layer. Total loss stands for the total network loss value calculated by log loss and loss box. For more details about the

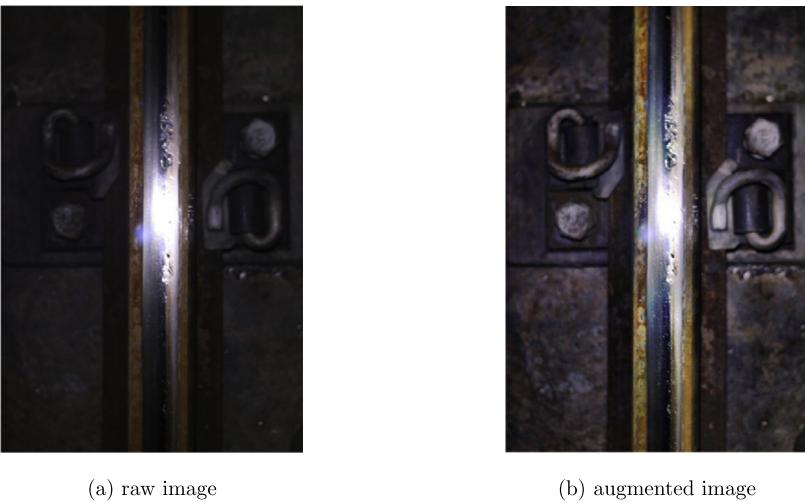


Fig. 24. Result of illumination augmentation.

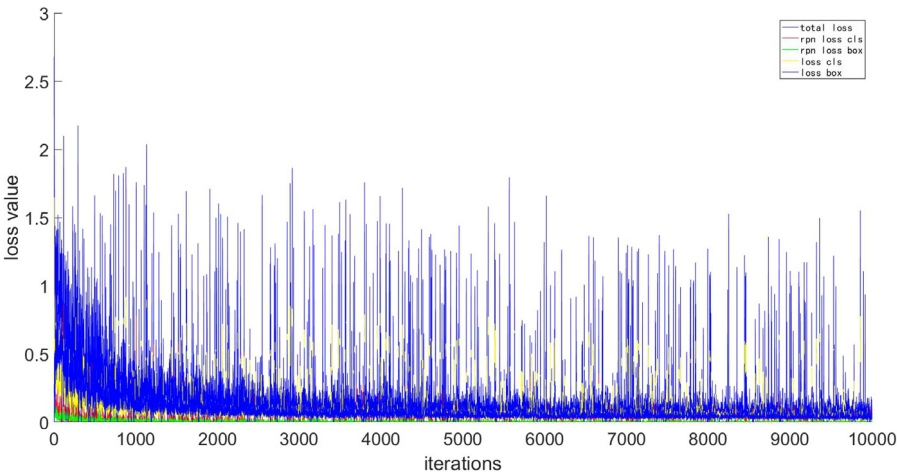


Fig. 25. The graph of the loss function of the Faster R-CNN.

Table 16
Detection rate comparison of the methods.

	Image processing + SVM	Image processing + CNN	Faster R-CNN
Positioning stage	2.17s	2.17s	0.23s
Classification stage	0.04s	0.03s	
Total	2.21s	2.20s	0.23s

definition and calculation of loss function, please refer [Ren et al. \(2015\)](#). According to the total loss values, after 5000 iteration, the network has converged. [Fig. 26](#) shows the boundary detected by Faster R-CNN.

[Table 13](#) provides the confusion matrix of fastener defect detection based on Faster R-CNN. There only 2 broken fasteners are misclassified to complete fastener. The precision of broken fastener is 100%, and the recall of broken fasteners is 98%. None of missing fasteners is misclassified. The precision and recall are 100%. However, there are 20 complete fasteners are not detected, which means the network could not find a bounding box that exceeded the set IOU threshold. In other words, a fastener is not detected which means the network recognizes it as background. The recall of complete fasteners is 96.06%. The total precision of fastener classification is 96.5%. The average-precision (AP) of each class and mean average precision(mAP) are the significant evaluation index of object detection. The value of AP is the area enclosed by the precision/recall curve. The mAP is the average of APs. The result of AP and mAP are given in [Table 14](#). The mAP of fastener detection is 97.9%.

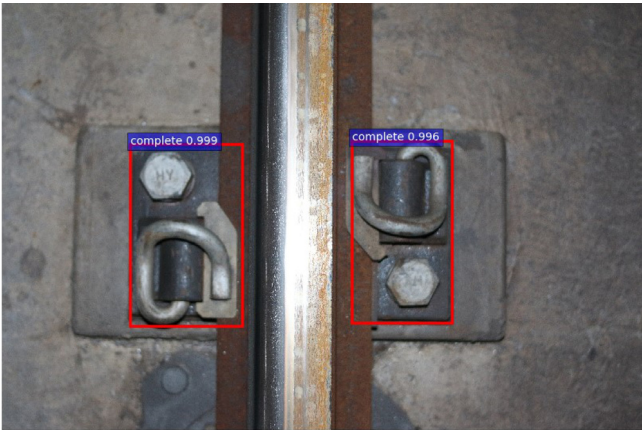


Fig. 26. The fastener boundary detected by Faster R-CNN.

Table 17
Detection accuracy of the methods.

	Dense SIFT + SVM	CNN	Faster R-CNN
Accuracy	99.26%	97.14%	97.90%

In addition to the detection accuracy of the algorithm, the time complexity and space complexity of the algorithm are also important indicators of the algorithm. Time complexity is the floating-point operations. Space complexity contains the number of weight and size of the feature map. The CNN computational cost is defined as follows.

$$Time \sim O(\sum_{l=h}^D M_l^2 \cdot K_l^2 \cdot C_{l-1} \cdot C_l) \quad (22)$$

$$Space \sim O(\sum_{l=1}^D K_l^2 \cdot C_{l-1} \cdot C_l + \sum_{l=1}^D M^2 \cdot C_l) \quad (23)$$

where D is the depth of network. l is the l th convolutional layer. K is the kernel length. C_l is the channel number of l th convolutional layer. M^2 is the size of feature map.

The computation cost of the algorithms used in this paper is shown in Table 15. As for the spatial complexity, the two networks are not much different. As the main parameter comes from the feature extraction network (VGG16). Faster R-CNN has more time complexity than VGG16 and these time complexity comes from the computation of RPN and RCNN.

Table 16 gives the time consuming for detecting every single image by the Faster-RCNN and the other methods presented in the paper. The time that Faster-RCNN uses to detect one single image is only 0.23s. This time is much shorter than the traditional fastener image detection method. The time spent by Faster R-CNN is only about 10% of the time spent by other methods. The time consuming of image processing in positioning stage is 2.17s, which is accounting for 98% of the total processing time. The reason is that wavelet transform and pattern matching are used in the positioning algorithm to improve the positioning accuracy, but it also result in high cost of time complexity. Faster R-CNN has a considerable advantage in the detection rate of target detection and classification.

Table 17 provides the fastener detection accuracy of the different methods used in this paper. Due to the increase of sample size, the accuracy of Faster R-CNN is increased compared with the accuracy of CNN. Although the accuracy does not exceed the accuracy of the method by using Dense SIFT and SVM, its advantages of fast and automatic positioning make Faster R-CNN based detection technology a better alternative for practical application.

7. Conclusions

In this paper, the railway track fastener defect (mainly for the fastener missing and broken cases) detection and classification issues are concerned. The method based on image processing, feature extraction and classifier are investigated in the first place. A robust and more precise positioning algorithm for the fastener is proposed. The Dense-SIFT, spatial pyramid decomposition and BOVW techniques are applied to the classification, and this method achieves a classification accuracy of 99.26%, which is better than the methods reported in the available literature. To simplify the classification procedure, a deep learning network based method is proposed in which the VGG16 network is used for the fastener defect recognition that achieves a sound classification accuracy of 97.14%. The DCNN method uses only one network and it is concise, compact, and easy to use than the image feature extraction based method. To the best knowledge of the authors, this method is the first one which introduces DCNN for fastener defect classification. Finally, to improve the recognition rate and consume less time for each image, the Faster R-CNN is proposed for the considered issues. This method simultaneously realizes the positioning and classification of the

fasteners. The mAP of fastener detection achieves 97.9% and the time of detection per image is 0.23s, which is almost ten times faster than the methods available in the literature.

All the methods proposed in this paper are demonstrated by the images collected from the Beijing Metro Line 6. It provides several alternatives for practical application. These methods will be applied to the field test in the near future, and it is part of our work in this project.

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